

Getting to know nano

One file, three outputs, zero dependencies

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2026-02-14

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A `.nano` file is a **research document**, not a program that happens to print text. Prose, code, data, figures and citations are the same document, so there is nothing to keep in sync and nothing to export by hand.

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The shape of a document

Anything you can write in Markdown works here, because Markdown is a subset. Two things are added: fenced code blocks that **run**, and directives that start with `@`.

```
let girth = [1.2, 1.6, 1.9, 2.3, 2.8, 3.1]
let mass  = [7.1, 12.4, 17.9, 24.2, 31.0, 38.4]

"n = {len(girth)}"
```

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```
n = 6
```

The value of the last expression in a cell is printed under it, and anything `print` output is captured too. The sample is small: mean girth 2.15 , and the coefficient of variation of mass is 0.536 .

NOTE

Cells share one scope, top to bottom, exactly like a notebook. A variable defined in cell 4 is available in cell 40, and in every `@plot` below it.

Charts are one block

A plot is a directive with `key = expression` lines. Data lives in the language, so you compute it, then draw it.

```
let t      = linspace(0, 12, 120)
let calm   = map(t, fn(x) => 1.6 * exp(-0.18 * x))
let noisy  = map(t, fn(x) => round(calm[min(floor(x * 10), len(calm) - 1)] * (1 + 0.12 * ...
```

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(no output)

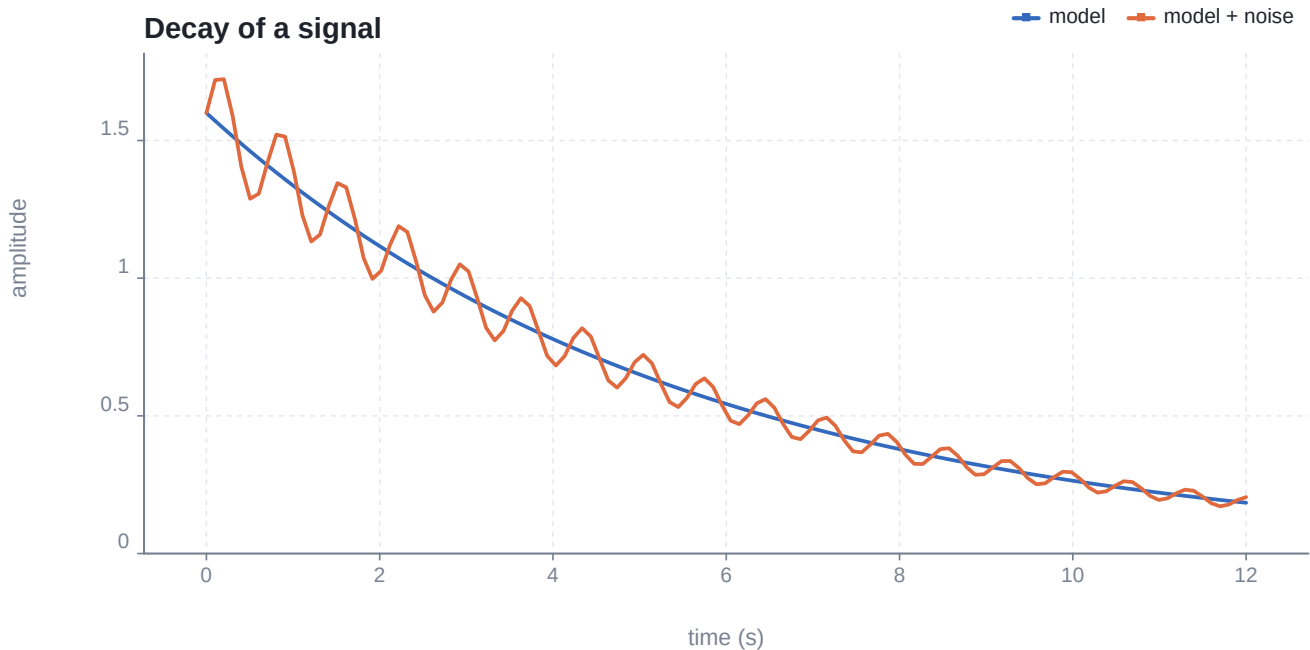


Figure 1. An exponential model with additive structure. Both series are computed in the cell above.

Every chart type works: `line`, `scatter`, `area`, `step`, `bar`, `hbar`, `hist`, `box`, `heatmap`, `errorbar`, `function`, `stem`, `density`.

```
seed(7)
let groups = ["control", "low dose", "high dose"]
let values = [randn(22), map(randn(22), fn(v) => v + 0.6), map(randn(22), fn(v) => v + 1...
```

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(no output)

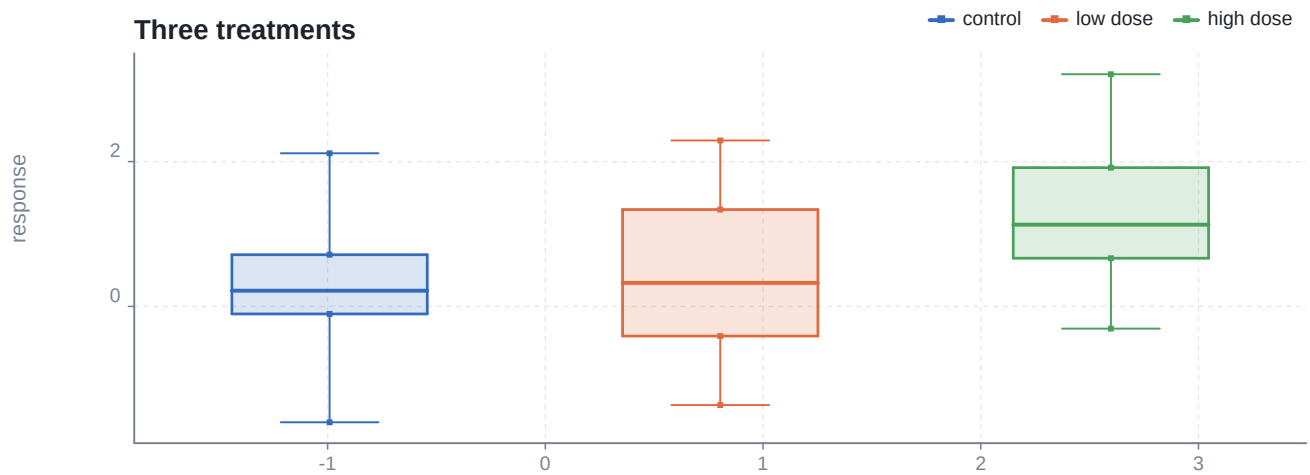
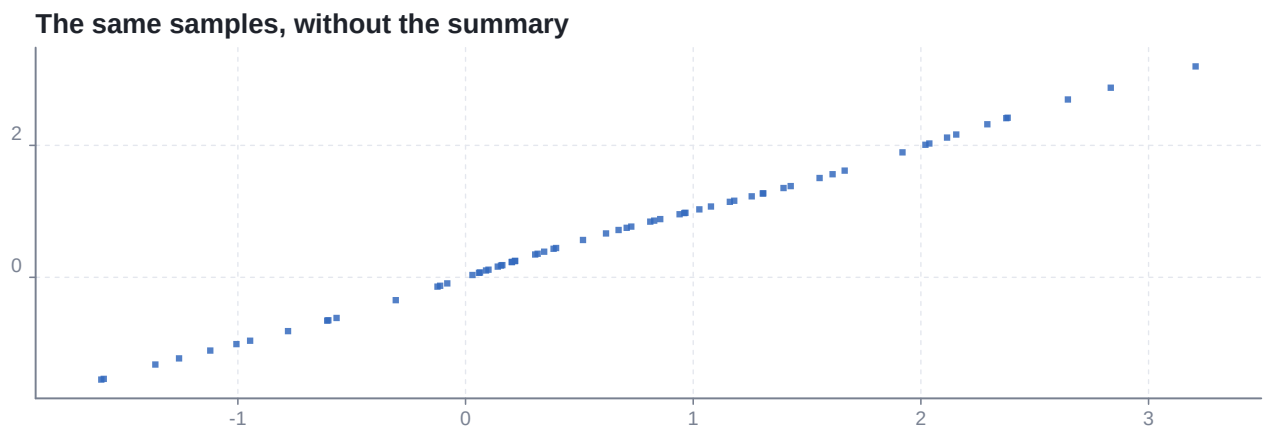


Figure 2. Box plots of 22 samples per arm.



Functions without ceremony

```
fn lorentz(x, gamma) {
  gamma / (pi * (x * x + gamma * gamma))
}
```

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```
let half_width = lorentz(_, 0.7)
let xs = linspace(-6, 6, 400)
let ys = map(xs, half_width)
```

```
"peak = {round(max(ys), 3)}, near x = {round(xs[index_of(ys, max(ys))], 2)}"
```

```
peak = 0.455, near x = -0.02
```

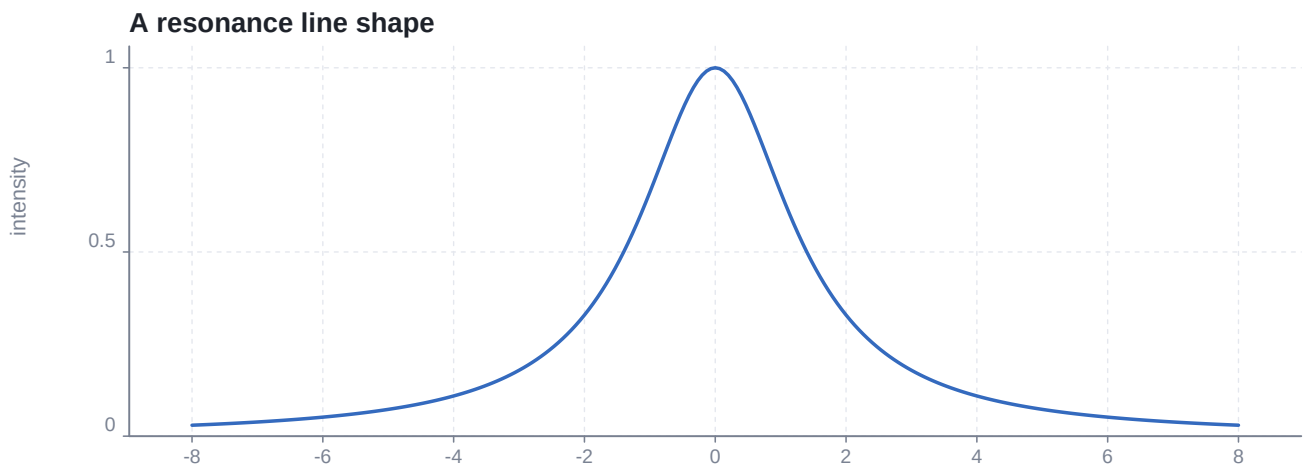


Figure 3. Plot formulas are evaluated by a small expression language that ships with the document.

WARNING

Formula plots can be made interactive. Declare a widget, bind it, and the figure redraws live in HTML while Markdown and PDF keep the initial value.

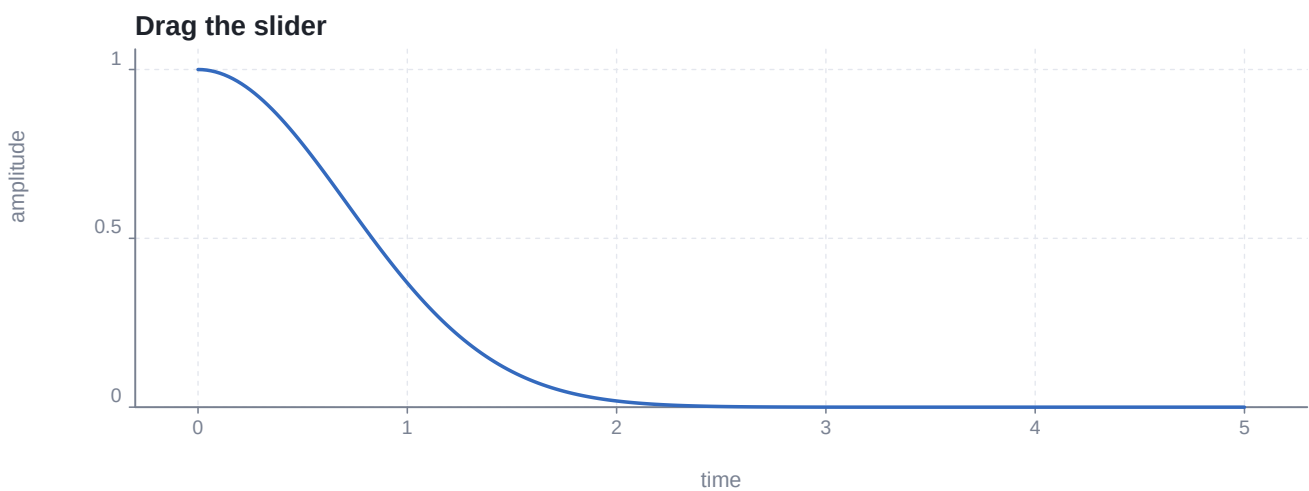
Interactive figures

damping rate



t

0.9 [0.1 .. 3] step 0.05



Current damping rate: 0.9

Data in tables

Table 1. The same numbers, in a table.

sample	girth	mass	ratio
1	1.2	7.1	5.92

sample	girth	mass	ratio
2	1.6	12.4	7.75
3	1.9	17.9	9.42
4	2.3	24.2	10.52
5	2.8	31	11.07
6	3.1	38.4	12.39

Records and grouping work too:

```
let rows = [
  {site: "north", temp: 4.1, n: 88},
  {site: "north", temp: 6.3, n: 91},
  {site: "south", temp: 19.8, n: 74},
  {site: "south", temp: 21.2, n: 70}
]

let by_site = group_by(rows, fn(r) => r.site)
"mean temp, south = {round(mean(map(by_site.south, fn(r) => r.temp))), 1})"
```

mean temp, south = 21

Table 2. Grouped by site.

site	n	mean temp
north	2	5.2
south	2	20.5

Math, notes and media

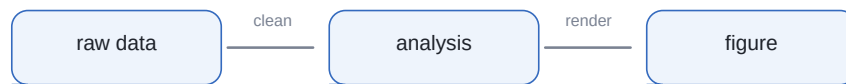
$$\hat{\beta}=(X^{top}X)^{-1}X^{top}y$$

Inline math such as $e^{i\pi}+1=0$ is typeset in the browser by a small built-in TeX renderer —no MathJax, no network, no build step. Display math works too:

$$\int_{-\infty}^{\infty} e^{-x^2} dx=\sqrt{\pi}$$

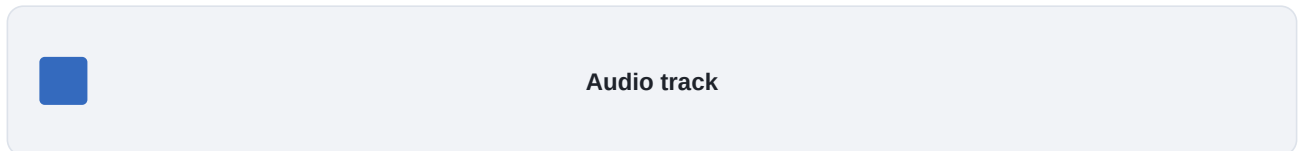
TIP

@note , @tip , @warning and @info all render as callout panels in every output format.



one .nano file in, three outputs out

Figure 4. Images are embedded in HTML and PDF, and copied alongside the Markdown.



Audio cannot be played in a PDF document

assets/tone.wav

Figure 5. A short tone (placeholder asset).



Embedded content cannot be displayed in a PDF document

<https://example.org/reference>

Figure 6. External reference, embedded as an `iframe` in HTML.

Where to go next

TIP

The full language reference lives in `docs/language.md`, and a task-oriented walkthrough lives in `docs/guide.md`. `nano build paper.nano -f pdf` writes one file; `-f all` writes Markdown, HTML and PDF side by side.